

Passive buzzer

DO

SOUND

PWM

What does it do?

A passive buzzer needs fast pulses (PWM) to make sound. Changing their frequency changes its pitch: low or high notes.

Recognise it: check for **passive**, **external drive** or **no oscillator** on the packet. Both types can look identical; + only marks polarity.

Use a small **3.3 V GPIO-compatible piezo** sounder.

Uses

Washing-machine chime

PWM plays tunes; active buzzers have only one pitch.



DETECT
Read cycle status



DECIDE
Has the wash finished?



DO
Play a short tune

Robot voice box

Change pitch for happy chirps or grumpy growls.



DETECT
Read mood button



DECIDE
Is happy pressed?



DO
Chirp high or low



+ leg
Marked +, to
GPIO

- leg
Other leg, to
GND

Vent
Sound comes
out

Code

● Raspberry Pi

```
from gpiozero import TonalBuzzer  
  
buzzer = TonalBuzzer(12)  
  
buzzer.play("A5")
```

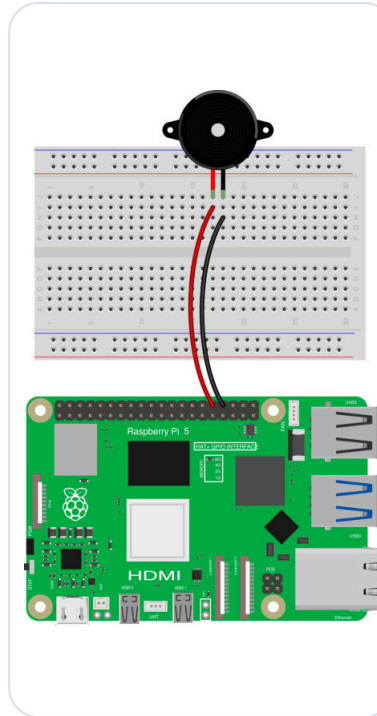
● Raspberry Pi Pico

```
from picozero import PWMBuzzer  
  
# GP16, 2000 Hz, 50% duty  
buzzer = PWMBuzzer(16, 2000, 32768)  
  
buzzer.on()
```

Pi: try `buzzer.play("A4")`; silence with `buzzer.stop()`. Pico: try `buzzer.freq = 1000`; silence with `buzzer.off()`. Higher frequency means higher pitch. Beep speed is separate; both types can make rhythms.

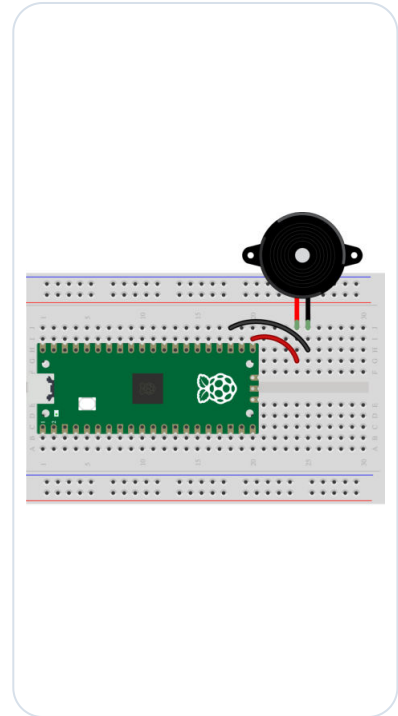
Wiring

RASPBERRY PI 5



GPIO 12 → + buzzer - → GND

PICO



GP16 → + buzzer - → GND